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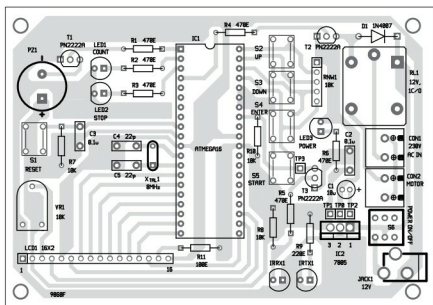


Fig. 4: Component layout of the PCB

circuit through JACK1 to provide the power supply. LED3 glows to indicate that the power has been connected. 230V AC mains is connected to CON1 and the motor is connected to CON2.

The operation of the circuit is as explained below:

1. When the circuit is switched on, the LCD shows the message as 'set object count.' Vary VR1 to control the contrast of the LCD if the message is not visible. (Initially, LEDs, buzzer and relay would be switched off.)

2. The user can set the desired count by incrementing or decrementing it by pressing Up/Down (S2/S3), respectively.

3. After the count is set, the user has to press Enter (S4). On pressing Enter, the LCD shows the message as 'count is set to xxxx' for two seconds and then next message is displayed as 'count up or down?' By default, count up is selected.

4. To change the counting direction, press Down (S3) for count down or Up (S2) for count up, followed by pressing Enter. After two seconds, the LCD shows next message as 'press start.'

5. When Start (S5) is pressed, the LCD shows the counting process. Then, as objects pass across the sen-

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

sor pair, the count is incremented or decremented while LED (LED1) is on to indicate that counting is in progress. A beep sound is produced through buzzer (PZ1) for every count. The relay (RL1) is energised and so the motor is on, pulling the conveyor belt till the selected number of objects is reached.

6. When the desired number of objects have been counted, the counting process stops. Green LED turns off and red LED (LED2) turns on to indicate that the counting process is over. The relay (RL1) is switched off to turn off the motor.

7. To repeat the process, reset the counter by pressing Reset (S1).

For hardware debugging, ensure various voltages are as given in the test points table. ●



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